Predictive Modelling of Heart Disease Risk Using Health Conditions and Behavioural Factors

Group Members

Rohitha Sai Kothapalli (11062657)

Nagarjuna Reddy Sreerangapu (11062445)

Avinash Ganisetti (11062199)

Under the guidance of

Prof. Reza Maihami

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Introduction

Cardiovascular diseases (CVDs) remain the leading cause of death worldwide, with nearly 18 million deaths reported annually. Despite progress in medical care, early detection of individuals at risk remains a major challenge. Traditional risk scoring methods often rely on generalized assumptions, failing to capture personal health differences and lifestyle factors highlighting the need for more dynamic, data-driven solutions.

Machine learning offers a powerful alternative by analyzing complex health data, including behavioral, clinical, and mental health variables. These models can identify hidden patterns and provide more accurate, personalized predictions. In doing so, they support early interventions, help patients take proactive steps, and enhance the effectiveness of preventive healthcare strategies.

Problem Statement

"Build a predictive model to assess heart disease risk using health conditions and behavioral factors."

This project seeks to develop a predictive model utilizing machine learning techniques to evaluate the risk of heart disease by analysing a combination of clinical health indicators, behavioural patterns, and mental health factors. By integrating these multidimensional aspects, the model aims to provide more precise and personalized risk assessments, thereby facilitating earlier detection, targeted preventive strategies, and improved long-term health outcomes.

Data Dictionary

The dataset utilized for this project consists of approximately 140,000 records, capturing a diverse range of demographic, behavioural, and health-related variables. It was obtained from Kaggle, a reputable online platform that provides datasets for data science and machine learning research. The data includes features such as age, body mass index (BMI), smoking behaviour, alcohol consumption, physical activity, and mental and physical health status. The target variable for prediction was a binary indicator of heart disease status (Yes/No), allowing for a classification-based modelling approach.

To ensure the data was suitable for analysis, a thorough preprocessing phase was conducted. This involved handling missing values, removing outliers, encoding categorical variables, and standardizing numerical fields. These steps were essential to enhance data quality and ensure robust and reliable model performance.

The dataset used in this study was obtained from a publicly available source on Kaggle and can be accessed at the following link:

https://www.kaggle.com/datasets/kamilpytlak/personal-key-indicators-of-heart-disease?resource=download

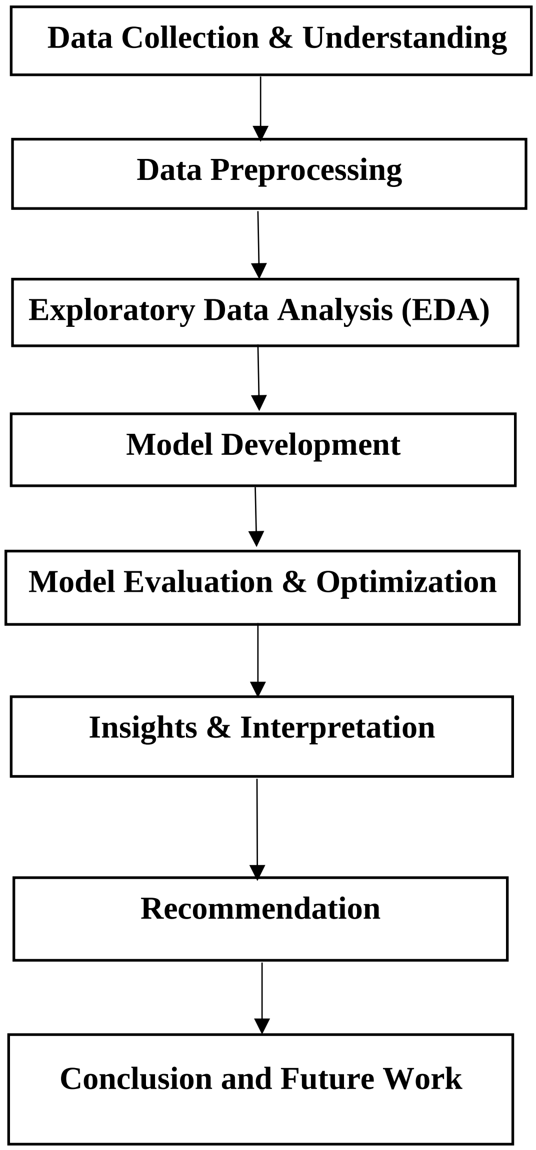
Column Name	Description
Heart Disease	Whether the individual has heart disease (Yes/No).
BMI	Body Mass Index (BMI) of the individual.
Smoking	Whether the individual has smoked at least 100 cigarettes in their lifetime (Yes/No).
Alcohol Drinking	Whether the individual is a heavy drinker (Yes/No).
Stroke	Whether the individual has had a stroke (Yes/No).
Physical Health	Number of days in the past 30 days the individual had poor physical health.
Mental Health	Number of days in the past 30 days the individual had poor mental health.
Diff Walking	Whether the individual has difficulty walking or climbing stairs (Yes/No).
Sex	Gender of the individual (Male/Female).
Age Category	Age group of the individual (e.g., 55-59, 80 or older, etc.).
Race	Race of the individual (e.g., White, Black, etc.).
Diabetic	Whether the individual has diabetes (Yes/No/Borderline).
Physical Activity	Whether the individual engages in physical activity other than work (Yes/No).
Gen Health	Self-reported general health status (Excellent, very good, Good, Fair, Poor).
Sleep Time	Average number of hours of sleep per day.
Asthma	Whether the individual has asthma (Yes/No).
Kidney Disease	Whether the individual has kidney disease (Yes/No).

Process flow

The overall process followed in this project is outlined in the flow diagram above, which represents a systematic approach to building an effective machine learning model for heart disease prediction.

- **Data Collection & Understanding:** The first step involved acquiring the dataset and performing an initial review to understand its structure, contents, and relevance to the research objective.
- **Data Preprocessing:** This phase included cleaning the dataset by handling missing values, removing duplicates and outliers, converting categorical variables, and standardizing data to ensure quality and consistency for analysis.
- **Exploratory Data Analysis (EDA):** EDA was conducted to uncover patterns, distributions, and potential relationships among variables. This step provided critical insights into variable behavior and informed the feature selection process.

- **Model Development:** Based on the EDA findings, multiple machine learning models—including Logistic Regression, Random Forest, XGBoost, and Neural Networks—were developed using training data.
- **Model Evaluation & Optimization:** The models were evaluated using key performance metrics such as accuracy, sensitivity, specificity, and AUC. Hyperparameter tuning and optimization were applied to enhance model performance.
- **Insights & Interpretation:** After evaluation, insights were drawn from the best-performing models to understand key risk factors and the predictive power of different variables.
- **Recommendation:** Based on the model's predictive outcomes, data-driven recommendations were formulated to support preventive strategies and early health interventions.
- **Conclusion:** Finally, the project outcomes were summarized, and potential future directions were identified to enhance prediction accuracy and application in real-world healthcare settings.



Literature Review

- Machine Learning Approaches for Heart Disease Prediction

Heart disease remains one of the leading causes of mortality worldwide, highlighting the need for advanced predictive models that facilitate early detection and intervention. Traditional diagnostic approaches rely on subjective assessments, which can lead to inconsistencies in evaluation. Machine learning (ML) has emerged as a crucial tool in automating and enhancing prediction accuracy, providing clinicians with valuable insights for risk assessment. This review explores various ML-based approaches and their effectiveness in predicting heart disease.

- Machine Learning Algorithms in Heart Disease Prediction

Several studies have utilized ML techniques to predict heart disease risk. A study by Ingole et al. (2024) evaluated multiple classifiers, including Logistic Regression, Random Forest, Decision Tree, Naïve Bayes, k-Nearest Neighbours, Neural Networks, and Support Vector Machine (SVM). The results indicated that SVM outperformed other models, achieving an accuracy of 91.51% (ResearchGate).

- Ensemble and Deep Learning Approaches

Ensemble learning techniques have shown promising results in heart disease prediction. A hybrid ensemble framework that integrates multiple ML models has demonstrated improved cardiovascular disease prediction accuracy (ResearchGate). Deep learning-based heart disease prediction models, such as artificial neural networks (ANNs), have also been effective in modeling complex relationships among risk factors (ESR Journal).

- Optimization Techniques in Heart Disease Risk Assessment

Optimization techniques enhance predictive model performance by refining feature selection, adjusting algorithm parameters, and reducing computational inefficiencies. A comprehensive review by Optimizing Cardiovascular Health discusses various strategies for primary prevention, highlighting the importance of early identification. Furthermore, the study on Heart Disease Prediction by Using Novel Optimization Algorithm demonstrates how optimization techniques improve predictive accuracy by leveraging major risk factors and refining classifier algorithms.

- Interpretative Strategies in Cardiovascular Risk Prediction

Interpretability in risk prediction is crucial for clinical adoption and decision-making. Several assessment tools provide transparency in determining heart disease risk. A literature review on cardiovascular risk assessment models evaluates their methodologies and applications, offering insights into their predictive value. Additionally, a narrative review of cardiovascular risk assessment scores discusses their role in quantifying risk, making them useful for both clinicians and patients.

- Financial Analyses and Socioeconomic Factors in Heart Disease Risk

Economic and social determinants significantly influence cardiovascular health. Financial stress has been linked to an increased risk of heart disease, as shown in a systematic review and meta-analysis that examines its association with coronary artery disease (CAD) and major cardiac events. Additionally, the integration of socioeconomic markers into risk models is emphasized in Social, Financial Factors Critical to Assessing Cardiovascular Risk, underscoring the importance of considering financial and social factors when evaluating an individual's risk.

- Challenges and Future Directions

Despite the success of ML models in heart disease prediction, several challenges remain. Data quality and availability continue to be major concerns, as biased or incomplete datasets can hinder model generalizability. Additionally, interpretability remains a crucial issue, particularly in deep learning models, where decision-making processes are often opaque. Future research should focus on integrating ML models with clinical decision support systems to enhance real-world applicability (PMC).

The reviewed studies highlight the effectiveness of ML techniques in predicting heart disease, with ensemble and deep learning methods demonstrating significant promise. Additionally, incorporating optimization strategies, interpretative techniques, and financial analyses into heart disease risk assessment provides a multidimensional approach that enhances predictive accuracy, clinical relevance, and real-world applicability. Advancements in feature selection, data preprocessing, and interpretability will further enhance the clinical utility of these models. The integration of ML-driven predictive analytics into healthcare systems can ultimately contribute to improved patient outcomes and early disease detection.

Data Cleaning

The cleaning process ensures that the dataset is properly structured before model training.

1. Handling Missing Data

The dataset was checked for missing values using sum(is.na(data)). Since no missing entries were detected, imputation techniques were not required, allowing the dataset to remain intact.

2. Data Type Conversion

Categorical variables such as *Smoking*, *AlcoholDrinking*, and *Stroke* were converted to factor types. Numerical variables like *BMI*, *PhysicalHealth*, *MentalHealth*, and *SleepTime* were converted to integer format for consistency.

3. Outlier Detection

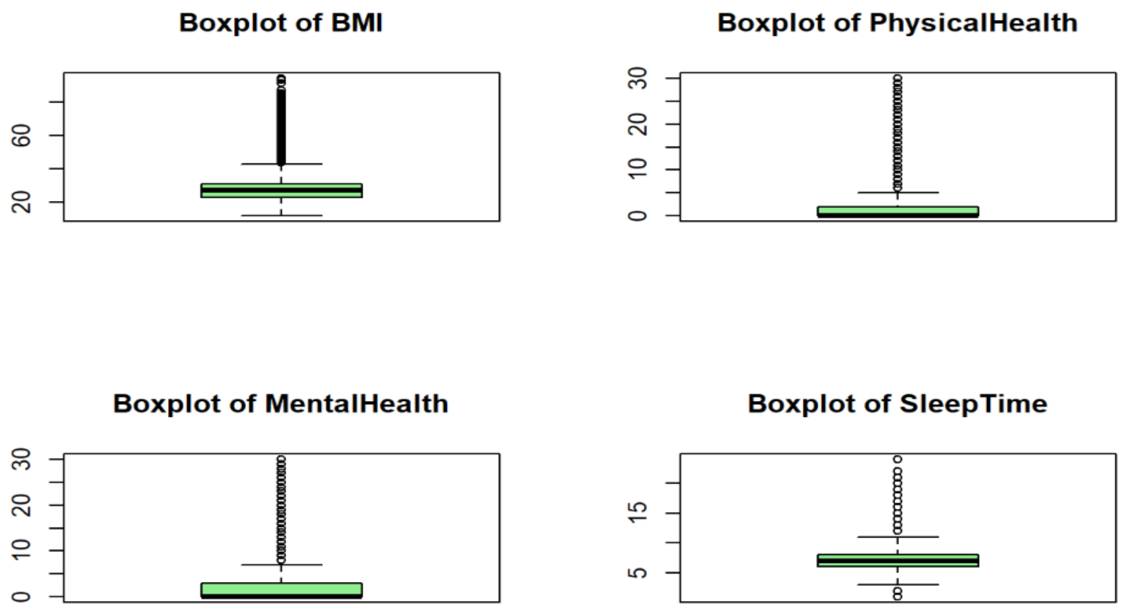
Boxplots were used to visually inspect outliers in *BMI*, *SleepTime*, *PhysicalHealth*, and *MentalHealth*. This helped identify extreme values that could potentially impact model performance.

Explanatory Data Analysis

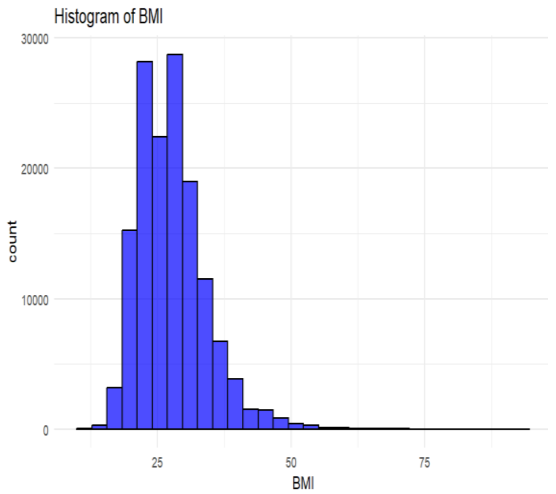
Exploratory Data Analysis (EDA) is a critical step in the data science process that involves summarizing and visualizing key characteristics of the dataset before formal modelling. It helps uncover underlying patterns, detect outliers or anomalies, examine relationships between variables, and assess the distribution of data. Using statistical summaries and graphical techniques: such as histograms, boxplots, and correlation heatmaps. EDA provides valuable insights that inform feature selection, guide preprocessing decisions, and shape the overall modelling approach.

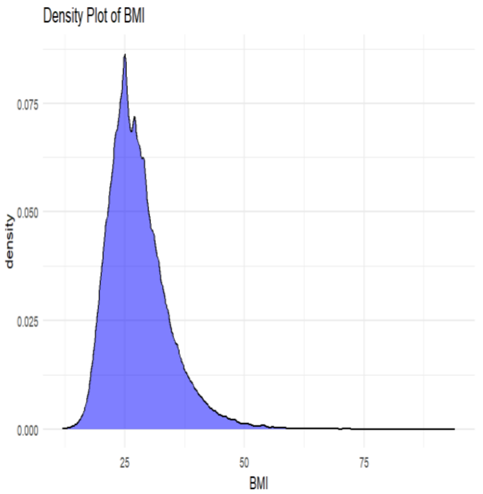
1. **For Numeric Variables**

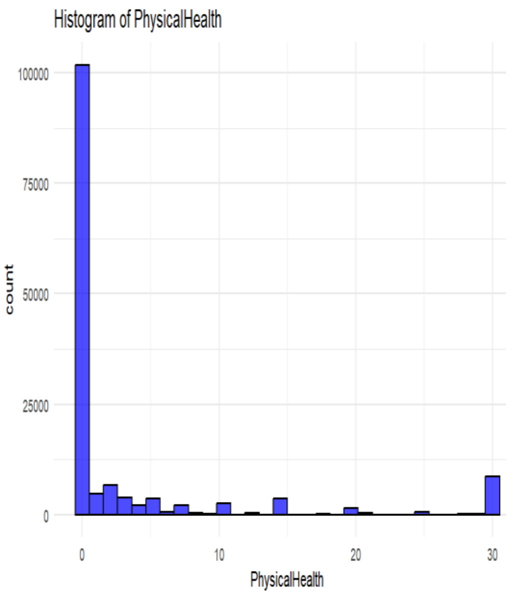
➔ Use density plots to visualize the probability distribution of BMI.

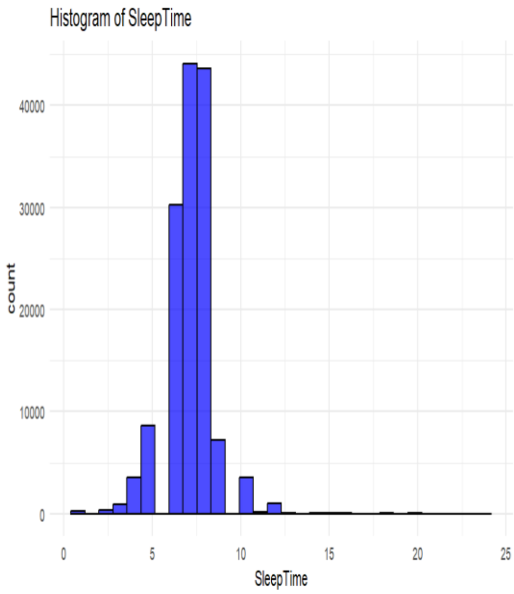


➔ Plot histograms to check the distribution (normal, skewed, etc.) BMI, SleepTime, and health metrics.

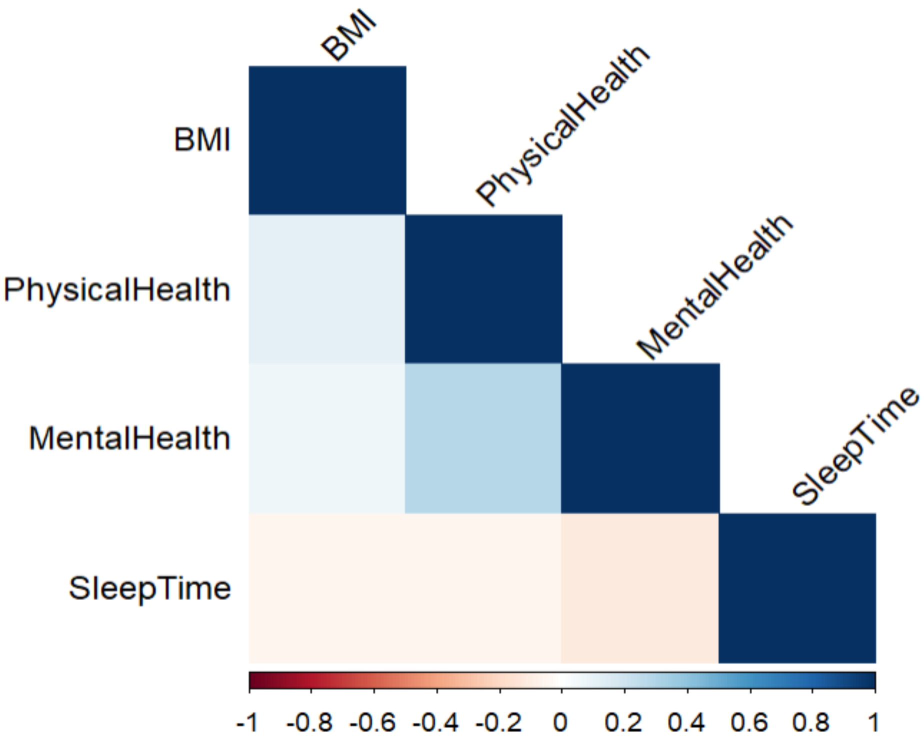




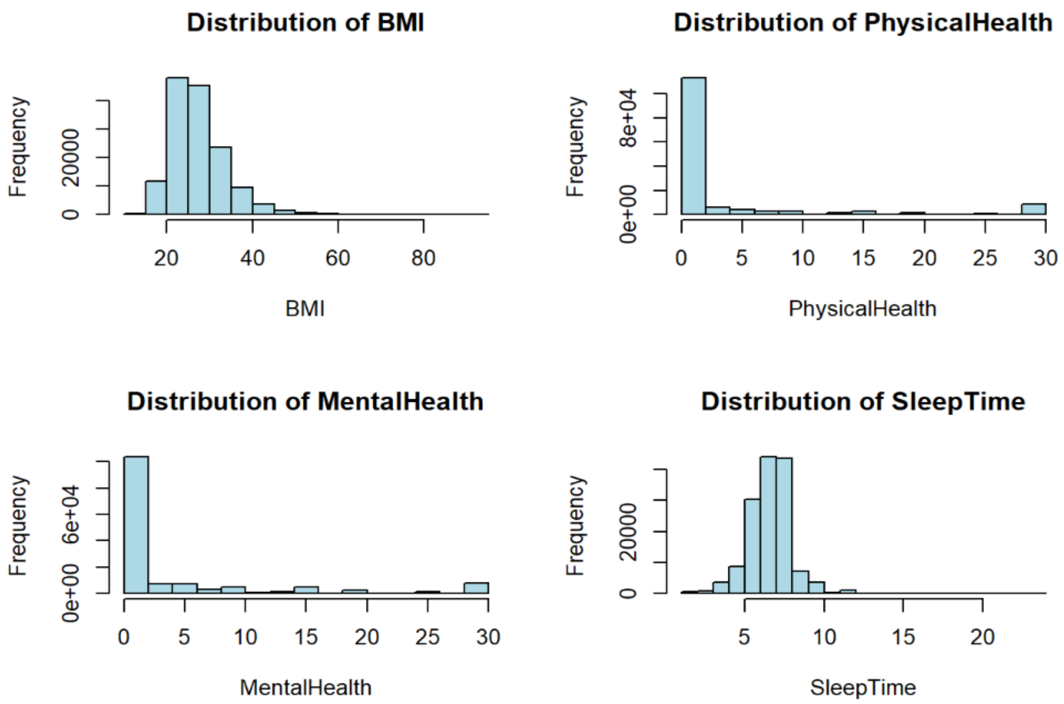




➔ Correlation Heatmap has displayed relationships between numerical features.



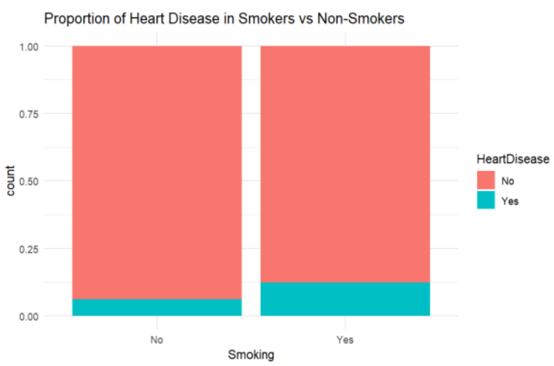
➔ Plot of Distribution of numeric variables (Histograms):

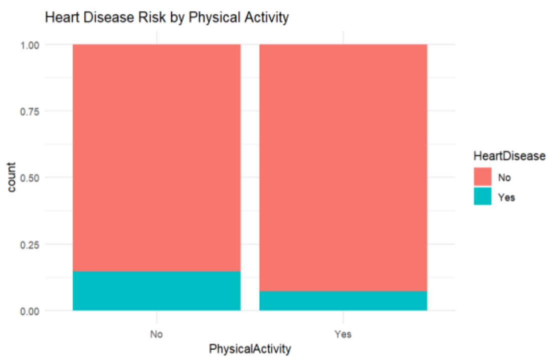


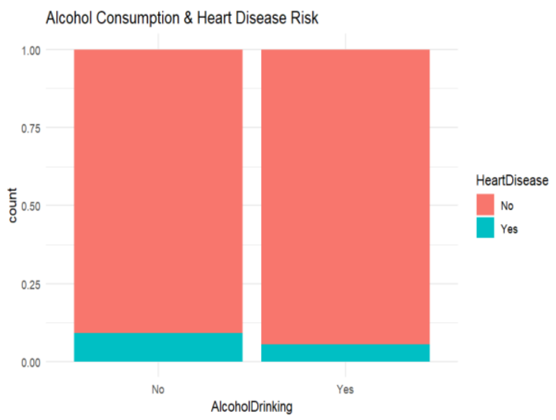
2. For Categorical Variables

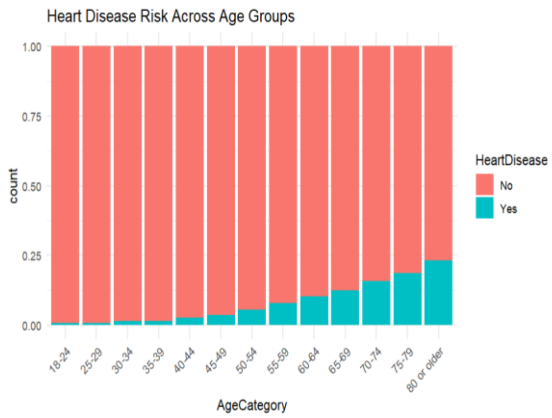
- Stacked Bar Charts used to show relationships between HeartDisease and categorical variables such as Smoking, AgeCategory. Compute proportions for percentage of smokers vs. non-smokers, etc.
-

➔ Few charts are represented below explaining the relationship between target variables and predictors.









Summary Statistics

- BMI, PhysicalHealth, and MentalHealth show high variation, indicating that extreme values or outliers should be handled carefully.
- A large proportion of people report 0 days of poor health, making it necessary to assess whether this variable provides meaningful variance.
- Smoking, diabetes, and poor general health are relatively common, reinforcing their importance as potential predictors of heart disease.
- SleepTime shows some extreme values, requiring further investigation.

Data Cleaning

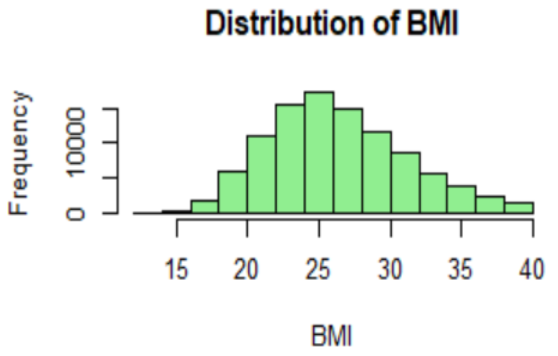
The data preprocessing phase involved several key steps to ensure the dataset was clean, consistent, and ready for modelling. Missing values were addressed either by removal or imputation where appropriate. Outliers were identified and removed using graphical techniques such as boxplots and histograms, ensuring they did not skew the results. Misclassified entries were corrected to maintain consistency across variables. Additionally, categorical variables were converted into dummy variables to make them compatible with machine learning algorithms. Relationships between variables were explored to validate data quality and ensure logical coherence. These steps collectively prepared the dataset for robust regression and classification modelling aimed at predicting heart disease risk.

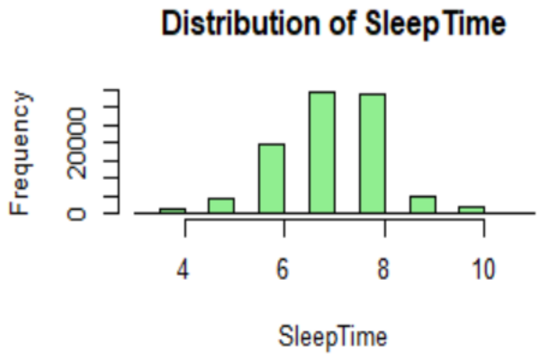
Handling Outliers in the Dataset

Outliers were detected and removed using the **Interquartile Range (IQR) method**, which is a robust statistical approach for identifying extreme values. This process was applied for only numerical variables in the dataset which are BMI and SleepTime. Ensuring that extreme values were successfully eliminated without affecting the core distribution of data.

After removing the outliers, the histograms of the above variables are plotted. The histograms provide a visual representation of the variable distributions, allowing for a better understanding of their spread and skewness. The dataset became cleaner and representative, leading to more accurate and reliable regression analysis.

- Histogram after removing the outlier: BMI and Sleep time





Model Development

For the modeling process, heart disease was designated as the target variable, while the remaining attributes served as predictors. The dataset was divided into an 80% training set and a 20% testing set to ensure unbiased model evaluation. To address class imbalance, resampling techniques were applied, improving the model's ability to learn from minority class instances. Numerical features were standardized using min-max normalization to bring them onto a common scale.

Multiple machine learning models including Logistic Regression, Decision Tree, Random Forest, XG-Boost, Neural Network, and Clustering were trained and tested. These models were then evaluated using performance metrics such as accuracy, sensitivity, specificity, and area under the ROC curve (AUC) to determine their effectiveness in predicting heart disease.

Data Partitioning

To ensure robust and unbiased model evaluation, the dataset was divided into two subsets: 80% for training and 20% for testing. The training set was used to teach the model underlying patterns, while the testing set evaluated its performance on new, unseen data. This partitioning strategy helps avoid overfitting and ensures that the model generalizes effectively to real-world scenarios. Stratified sampling was applied to preserve the proportion of the target variable in both sets, maintaining class balance. Additionally, a random seed was set to ensure consistency and reproducibility of results throughout the modelling process.

Balancing the Data

Initial exploration revealed a significant class imbalance, with a disproportionately higher number of individuals without heart disease compared to those with it. This imbalance can lead models to favor the majority class, resulting in biased predictions. To mitigate this, the ROSE (Random Over-Sampling Examples) technique was employed to generate a balanced dataset. This method synthetically adds new samples from the minority class while adjusting to the majority class. After applying ROSE, the dataset achieved an approximately equal distribution between positive and negative cases, allowing for more accurate learning and fairer classification.

Normalization of Data

The dataset included several numeric features—such as BMI, Physical Health, Mental Health, and Sleep Time—that varied widely in scale. Since many machine learning algorithms are sensitive to feature magnitude, normalization was necessary to ensure uniformity across variables. A min-max normalization technique was applied, which scaled all numeric values to a range between 0 and 1. This transformation not only stabilized the learning process but also improved model convergence and predictive accuracy.

Regression Models

1. Logistic Regression

Logistic Regression model was built to predict whether an individual has heart disease based on various health and lifestyle-related variables. This model is optimized for binary classification, making it well-suited for distinguishing between individuals with and without heart disease. Logistic regression is run using all the variables as predictors as the primary model for identifying which variables are significant and also to get the model outcome.

Key risk factors include smoking, diabetes, poor general health, higher BMI, and older age, all showing strong statistical significance. The model fit is strong, as indicated by a reduction in deviance and AIC = 50402. These results emphasize the importance of lifestyle and health factors in predicting heart disease.

The logistic regression model output reveals that a majority of the predictor variables are statistically significant in determining heart disease risk. Significance is determined by the p-value (Pr(>|z|)), with values less than 0.05 indicating that the variable has a meaningful impact on the model outcome.

➔ From the output:

- Over 90% of the variables have p-values < 0.05, many of which are highly significant with p-values < 0.001 (denoted by ***).
- Key predictors such as Smoking, Alcohol Consumption, Stroke History, Physical and Mental Health, BMI, and various Age Categories are all statistically significant.
- Health-related factors like Diabetes, Physical Activity, General Health, and Chronic Conditions (e.g., Asthma, Kidney Disease, Skin Cancer) also contribute significantly.
- The AIC value (108703) and reduction in deviance (from 153826 to 108627) indicate a well-fitting model.

These results demonstrate that lifestyle, age, and medical history variables strongly influence heart disease risk, validating their importance in predictive modeling.

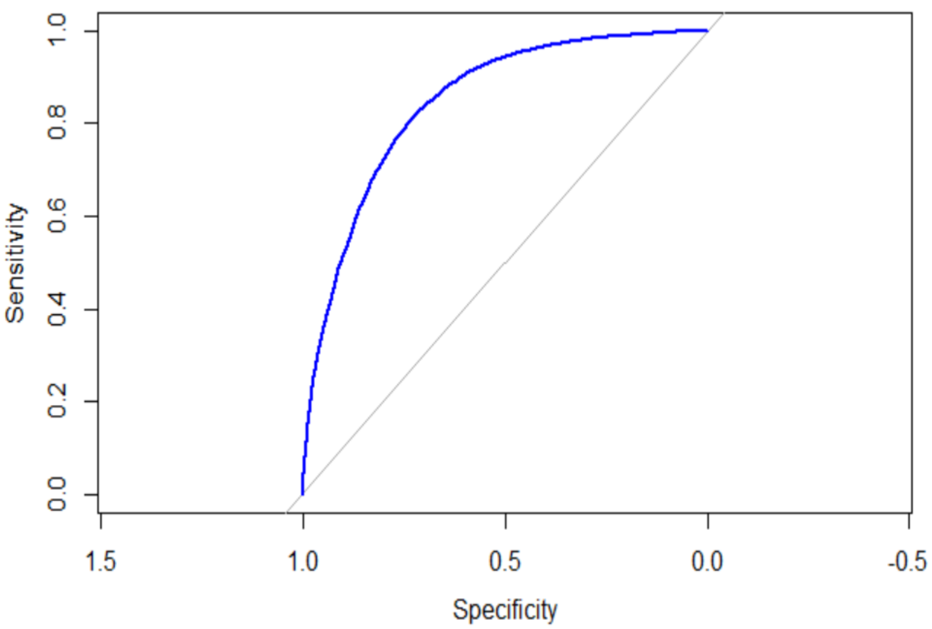
```
##
## Call:
## glm(formula = HeartDisease ~ ., family = binomial, data = train_data)
##
## Coefficients:
##                                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)                        -4.22990    0.10731 -39.419  < 2e-16 ***
## BMI                                 0.34491    0.05859   5.887 3.93e-09 ***
## SmokingYes                          0.35631    0.01539  23.151  < 2e-16 ***
## AlcoholDrinkingYes                 -0.19356    0.03274  -5.912 3.38e-09 ***
## StrokeYes                           1.23455    0.03285  37.585  < 2e-16 ***
## PhysicalHealth                      0.38005    0.06092   6.238 4.43e-10 ***
## MentalHealth                        0.31466    0.05258   5.984 2.17e-09 ***
## DiffWalkingYes                      0.27926    0.02156  12.956  < 2e-16 ***
## SexMale                             0.74411    0.01566  47.511  < 2e-16 ***
## AgeCategory25-29                    0.02709    0.09484   0.286 0.775132
## AgeCategory30-34                    0.54407    0.08311   6.546 5.91e-11 ***
## AgeCategory35-39                    0.46014    0.08260   5.571 2.53e-08 ***
## AgeCategory40-44                    0.92457    0.07666  12.061  < 2e-16 ***
## AgeCategory45-49                    1.21188    0.07433  16.303  < 2e-16 ***
## AgeCategory50-54                    1.65381    0.07121  23.224  < 2e-16 ***
## AgeCategory55-59                    1.98148    0.06986  28.363  < 2e-16 ***
## AgeCategory60-64                    2.21067    0.06911  31.989  < 2e-16 ***
## AgeCategory65-69                    2.56823    0.06880  37.327  < 2e-16 ***
## AgeCategory70-74                    2.83016    0.06900  41.014  < 2e-16 ***
## AgeCategory75-79                    3.00837    0.07014  42.892  < 2e-16 ***
## AgeCategory80 or older              3.34865    0.06990  47.906  < 2e-16 ***
## RaceAsian                          -0.54403    0.08574  -6.345 2.22e-10 ***
## RaceBlack                          -0.24509    0.07047  -3.478 0.000506 ***
## RaceHispanic                        0.45833    0.07157   6.404 1.51e-10 ***
## RaceOther                          -0.07644    0.07662  -0.998 0.318478
## RaceWhite                          -0.06910    0.06536  -1.057 0.290409
## DiabeticNo, borderline diabetes     0.15133    0.04577   3.306 0.000946 ***
## DiabeticYes                         0.51895    0.02001  25.939  < 2e-16 ***
## DiabeticYes (during pregnancy)      0.24146    0.09764   2.473 0.013398 *
## PhysicalActivityYes                 0.01341    0.01817   0.738 0.460417
## GenHealthFair                       1.63848    0.03203  51.150  < 2e-16 ***
## GenHealthGood                       1.10617    0.02628  42.092  < 2e-16 ***
## GenHealthPoor                       2.02586    0.04716  42.957  < 2e-16 ***
## GenHealthVery good                  0.52445    0.02622  20.004  < 2e-16 ***
## SleepTime                          -0.38069    0.05972  -6.374 1.84e-10 ***
## AsthmaYes                           0.31168    0.02206  14.128  < 2e-16 ***
## KidneyDiseaseYes                    0.61986    0.03252  19.060  < 2e-16 ***
## SkinCancerYes                       0.16843    0.02224   7.572 3.67e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 153826  on 110962  degrees of freedom
## Residual deviance: 108627  on 110925  degrees of freedom
## AIC: 108703
##
## Number of Fisher Scoring iterations: 5
```

After training the logistic regression model on the balanced and normalized dataset, I used it to predict heart disease outcomes on the test data. The model gave a prediction accuracy of 76.86%.

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    No   Yes
##        No  10440  2947
##        Yes  3472 10880
##
##                Accuracy : 0.7686
##                  95% CI : (0.7636, 0.7735)
##     No Information Rate : 0.5015
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.5372
##
##  Mcnemar's Test P-Value : 6.14e-11
##
##             Sensitivity : 0.7504
##             Specificity : 0.7869
##          Pos Pred Value : 0.7799
##          Neg Pred Value : 0.7581
##              Prevalence : 0.5015
##          Detection Rate : 0.3764
##    Detection Prevalence : 0.4826
##       Balanced Accuracy : 0.7686
##
##        'Positive' Class : No
##
```

➔ Sensitivity: 75.04% – the model was good at detecting people without heart disease.
➔ Specificity: 78.69% – also good at identifying those with the condition.

We also plotted the which visually shows the trade-off between sensitivity and specificity. The Area Under the Curve (AUC) was 0.8448, which is quite good and suggests the model has strong discriminatory ability between the two classes.



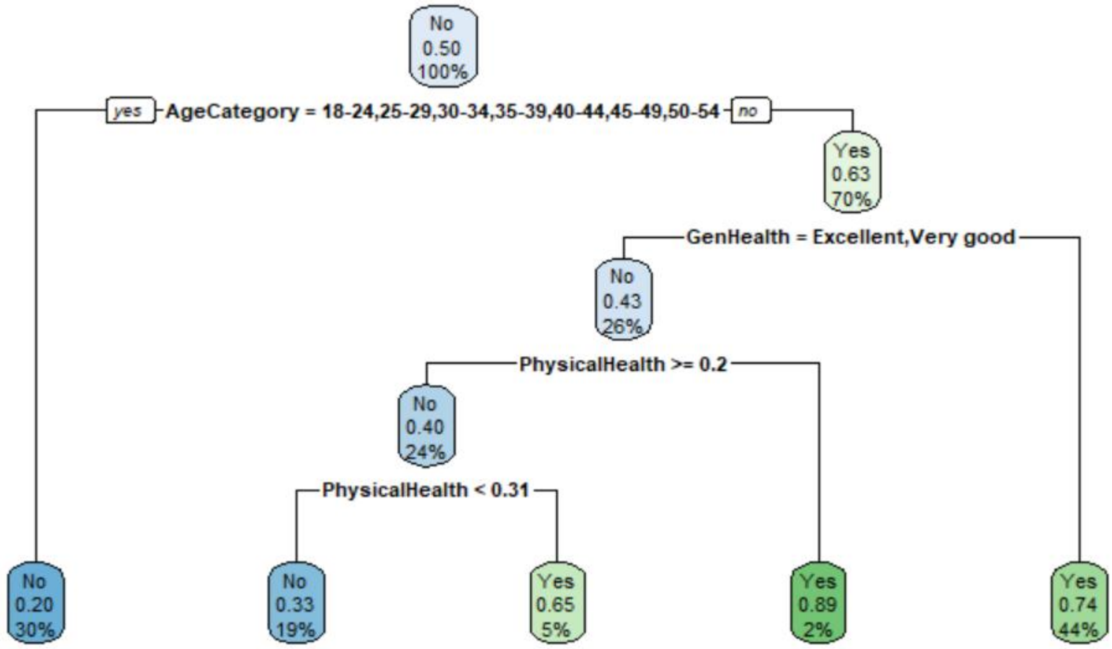
2. Decision Tree

A Decision Tree model was built to classify individuals as having heart disease or not, based on a set of health-related and behavioral predictors. The model works by learning simple decision rules from the data and is known for being easy to interpret and visualize. The model achieved a prediction accuracy of 74.23%.

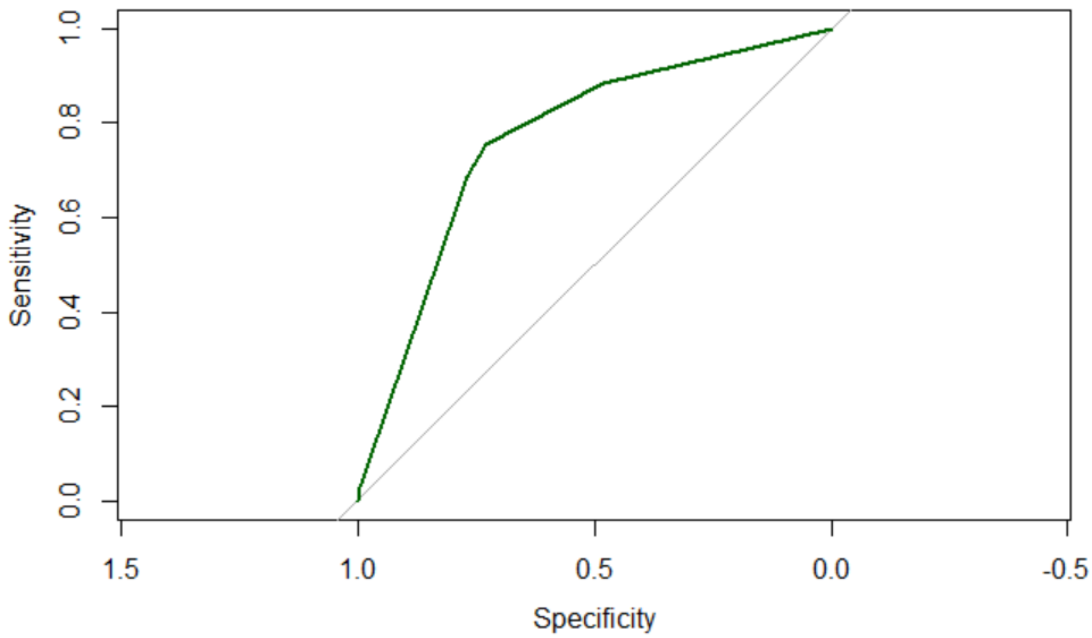
```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    No   Yes
##        No  10149  3385
##        Yes  3763 10442
##
##                Accuracy : 0.7423
##                  95% CI : (0.7371, 0.7475)
##     No Information Rate : 0.5015
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.4847
##
##  Mcnemar's Test P-Value : 8.23e-06
##
##             Sensitivity : 0.7295
##             Specificity : 0.7552
##          Pos Pred Value : 0.7499
##          Neg Pred Value : 0.7351
##              Prevalence : 0.5015
##          Detection Rate : 0.3659
##    Detection Prevalence : 0.4879
##       Balanced Accuracy : 0.7424
##
##        'Positive' Class : No
##
```

- ➔ Sensitivity: 72.95% – the model did a decent job identifying people without heart disease.
- ➔ Specificity: 75.52% – it also performed well in detecting those who actually had heart disease.

The decision tree visualizes how heart disease risk varies based on age, general health, and physical health. It shows that individuals aged 18–54 have a lower risk, especially if they report good general and physical health. For older age groups, the risk increases, particularly when general health is poor and physical health declines. This model helps identify key risk paths and highlights the importance of age and health perceptions in predicting heart disease.



We also plotted the ROC curve, which showed how the model balances between sensitivity and specificity. The Area Under the Curve (AUC) was 0.7671, which indicates a good ability to distinguish between the two classes. While not as powerful as some other models, the decision tree still provided a solid and interpretable approach to predicting heart disease.



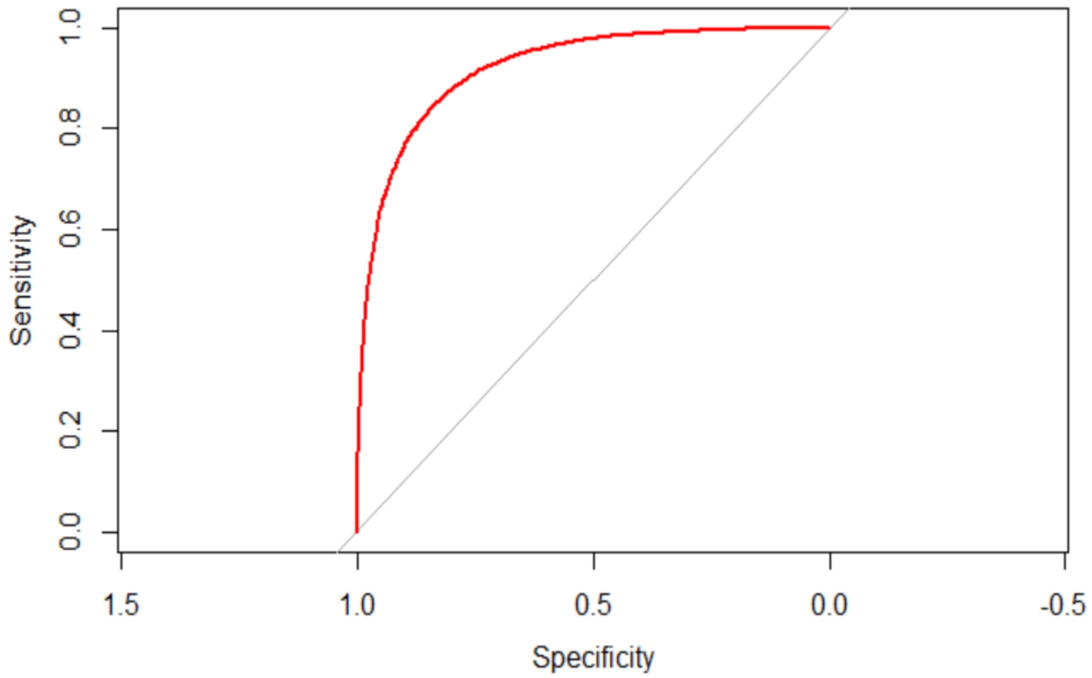
3. Random Forest

The Random Forest model was developed to improve prediction accuracy by combining multiple decision trees and averaging their results. It is known for its robustness and ability to handle complex relationships between variables in large datasets. The model performed very well, achieving an impressive accuracy of 83.99%.

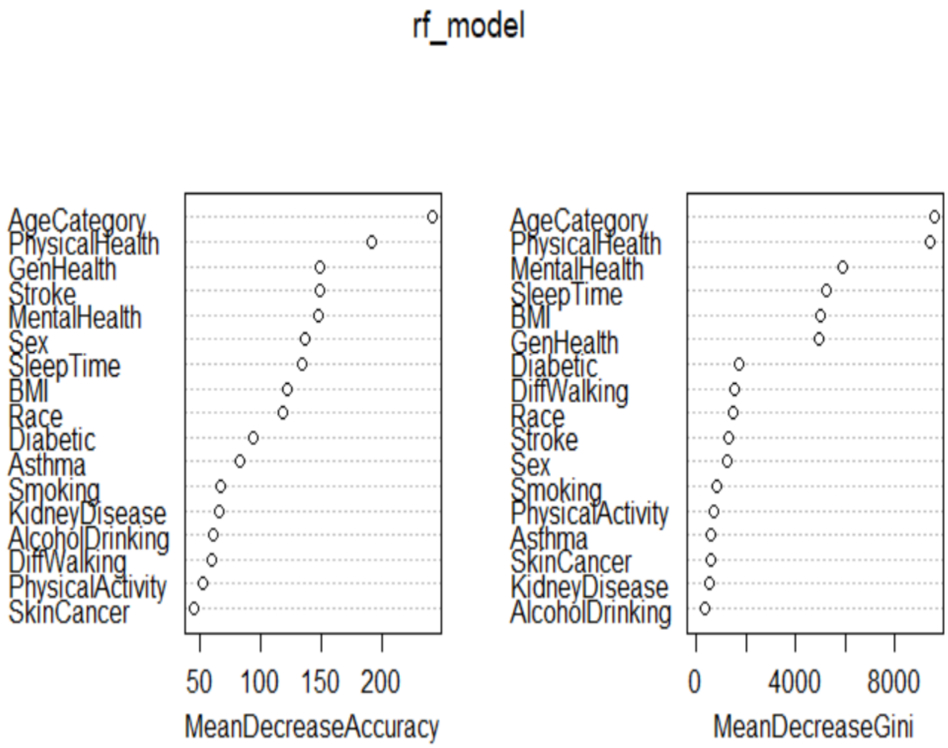
```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    No   Yes
##        No  11224  1753
##        Yes  2688 12074
##
##                Accuracy : 0.8399
##                  95% CI : (0.8355, 0.8442)
##     No Information Rate : 0.5015
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.6799
##
##  Mcnemar's Test P-Value : < 2.2e-16
##
##             Sensitivity : 0.8068
##             Specificity : 0.8732
##          Pos Pred Value : 0.8649
##          Neg Pred Value : 0.8179
##              Prevalence : 0.5015
##          Detection Rate : 0.4046
##    Detection Prevalence : 0.4678
##       Balanced Accuracy : 0.8400
##
##        'Positive' Class : No
##
```

- ➔ Sensitivity: 80.68% – the model was quite good at identifying people without heart disease.
- ➔ Specificity: 87.32% – it was even better at detecting those who had the disease.
- ➔ Kappa: 0.6799 – showing substantial agreement between actual and predicted values.

The ROC curve showed excellent model performance, with an AUC of 0.9211, indicating a strong ability to distinguish between the two classes.

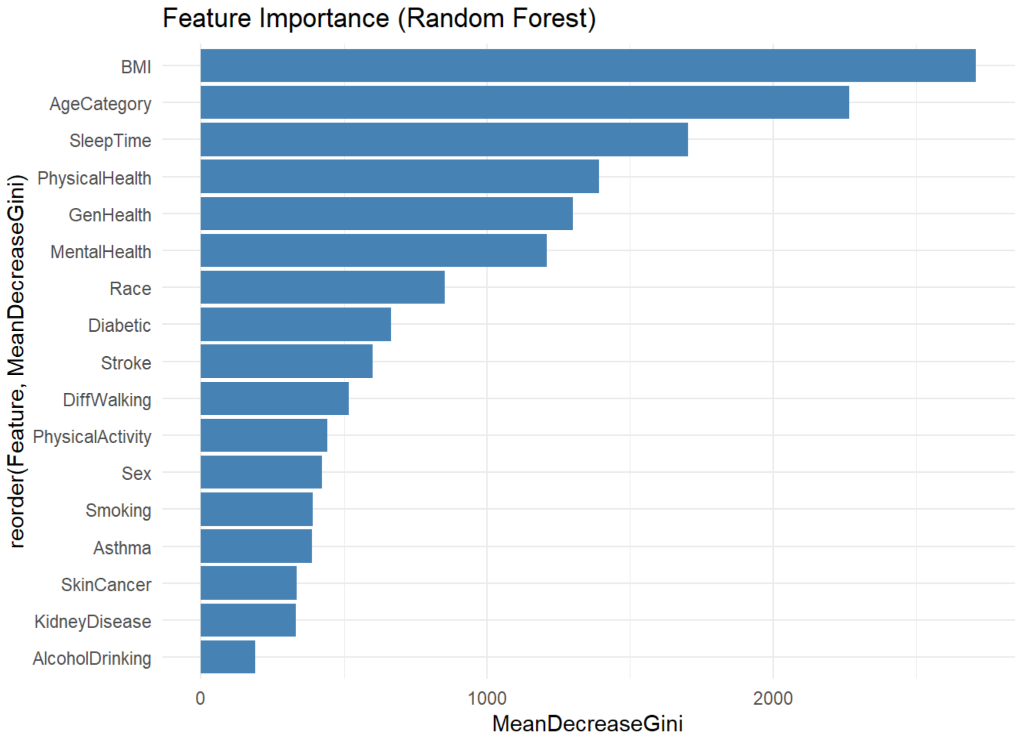


In addition, the feature importance values showed that the most influential variables were AgeCategory, PhysicalHealth, MentalHealth, SleepTime, and General Health. This means these factors played a major role in predicting the risk of heart disease. Overall, random forest was the most accurate and robust model among the ones tested in this project.



Feature Importance of Variables

Using the Random Forest model, BMI, Age Category, and Sleep Time were identified as the most important factors for predicting heart disease, while variables like Alcohol Drinking and Kidney Disease showed the least importance.



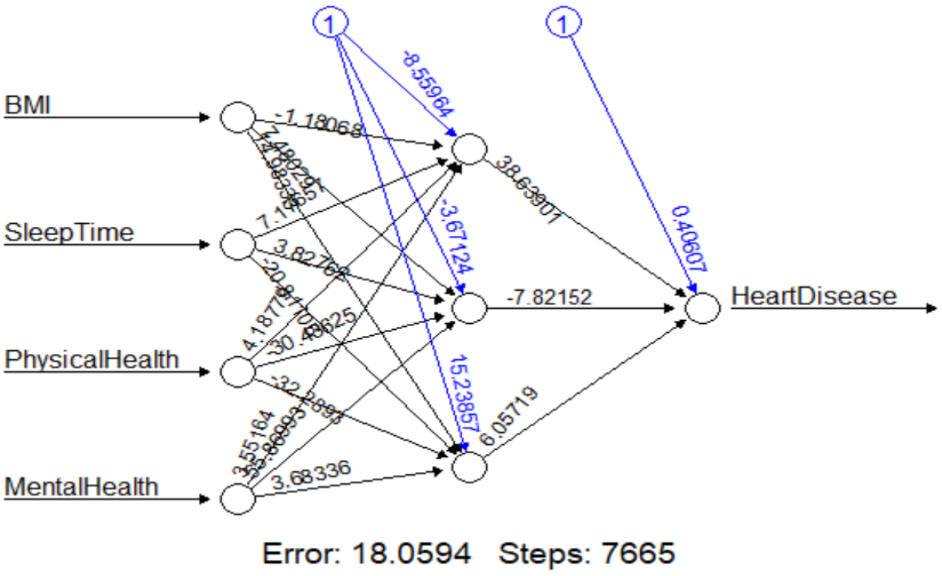
4. Neural Networks

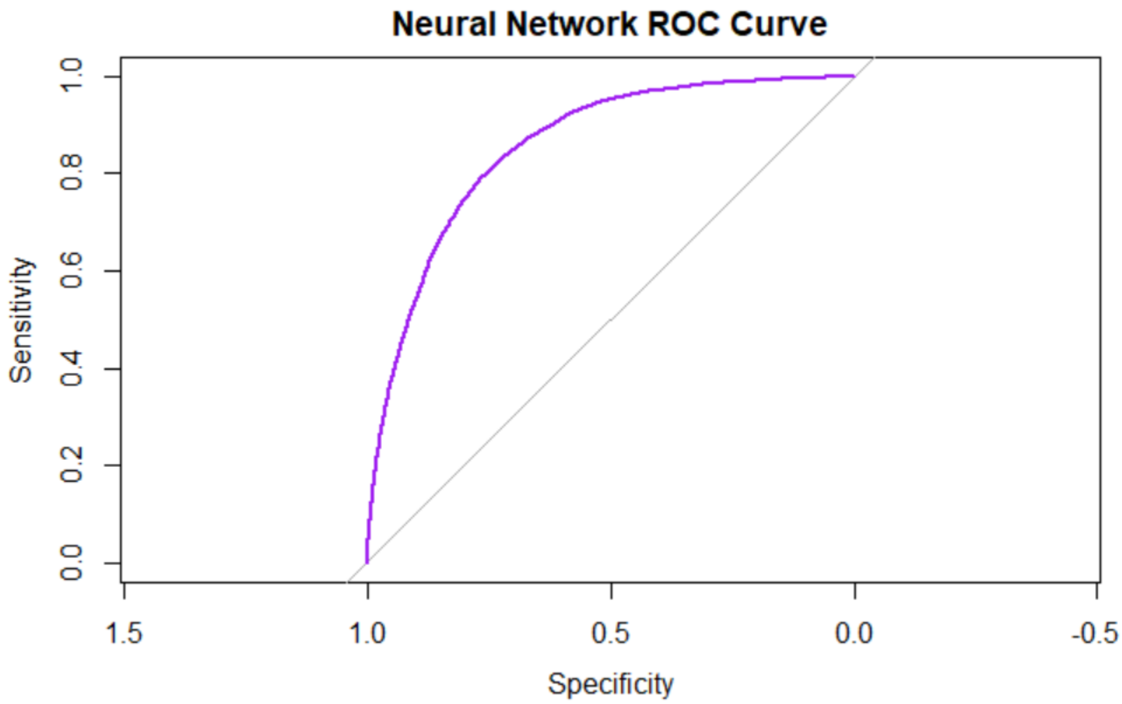
A Neural Network model was built to capture complex, non-linear patterns in the data for predicting heart disease. This model mimics the human brain's structure and is capable of learning intricate relationships between health and lifestyle variables. The model achieved a prediction accuracy of 77.74%.

➔ Sensitivity: 74.43% – the model did a good job identifying people without heart disease.
➔ Specificity: 81.06% – it also performed well in detecting those who had heart disease.

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction     0     1
##          0 10355  2619
##          1  3557 11208
##
##                Accuracy : 0.7774
##                  95% CI : (0.7724, 0.7822)
##     No Information Rate : 0.5015
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.5548
##
##  Mcnemar's Test P-Value : < 2.2e-16
##
##             Sensitivity : 0.7443
##             Specificity : 0.8106
##          Pos Pred Value : 0.7981
##          Neg Pred Value : 0.7591
##              Prevalence : 0.5015
##          Detection Rate : 0.3733
##    Detection Prevalence : 0.4677
##       Balanced Accuracy : 0.7775
##
##        'Positive' Class : 0
##
```

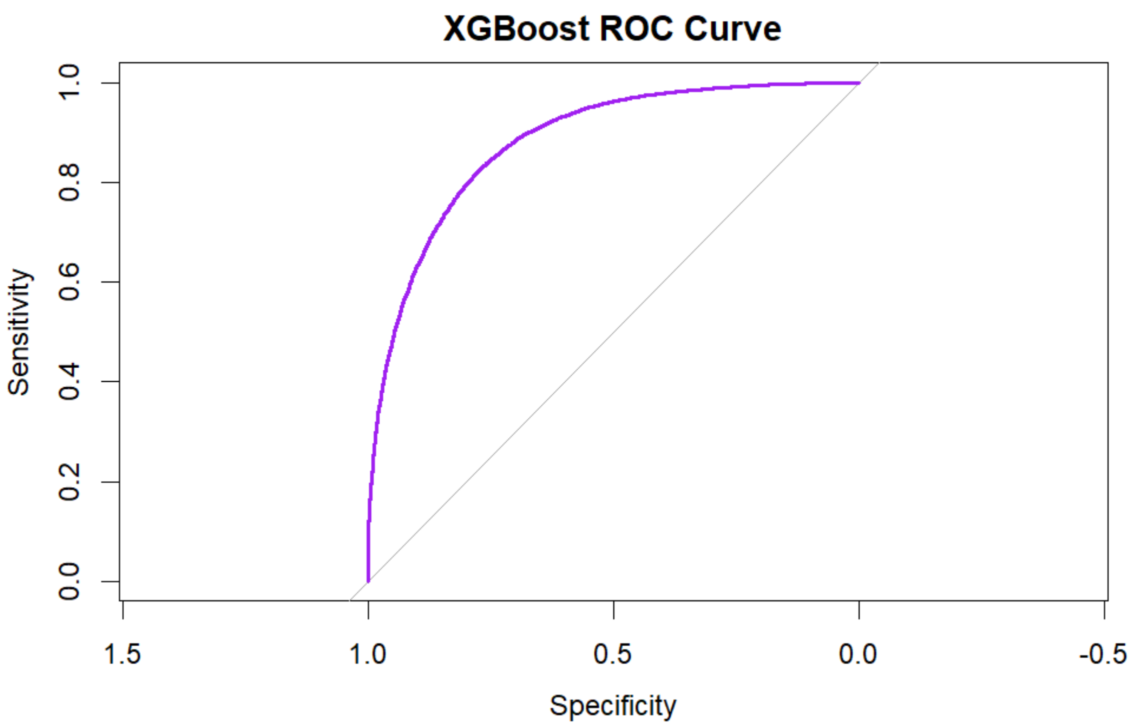
Plotted the ROC curve, which showed how the model balances between sensitivity and specificity. The Area Under the Curve (AUC) was 0.8547, indicating a strong ability to distinguish between the two classes. The neural network outperformed several other models in terms of accuracy and AUC, making it a robust choice for heart disease prediction, although it requires more computational resources and is less interpretable compared to simpler models.





5. XG-Boost Method

The XG-Boost model was implemented to enhance prediction efficiency and accuracy using gradient boosting, which combines multiple weak learners to form a strong predictive model. It is known for its speed, performance, and scalability. After testing the model on unseen data, it achieved an accuracy of 79.7%, which was one of the best results among all the models.



- ➔ Sensitivity: 79.3% – the model was good at correctly identifying people who had heart disease.
- ➔ Specificity: 80.0% – it also did a solid job recognizing those without the disease.

To evaluate the model further, I plotted the ROC curve, which helped show how well it balances sensitivity and specificity. The AUC (Area Under the Curve) came out to be 0.881, indicating that the model is quite strong.

Overall, XG-Boost outperformed several other models in both accuracy and AUC. It turned out to be a reliable and efficient option for this task. While it's more complex than simpler models like logistic regression, it still runs quickly and doesn't require as much computational power as a neural network, making it a practical choice in many situations.

Performance Evaluation

To evaluate and compare the performance of all models used in this study, looking at key metrics including accuracy, sensitivity, specificity, and AUC. These comparisons helped identify which models are not only statistically reliable but also practically useful for predicting heart disease. From the above performed models for logistic model, decision tree, random forest model, neural networks and XG-Boost models identifying the best model.

Regression Models	Sensitivity	Specificity	Accuracy	AUC
Logistic	0.750	0.787	0.769	0.845
Decision Tree	0.729	0.755	0.742	0.767
Random Forest	**0.807**	**0.873**	**0.839**	**0.921**
Neural Networks	0.744	0.811	0.777	0.855
XG-Boost	0.793	0.800	0.797	0.881

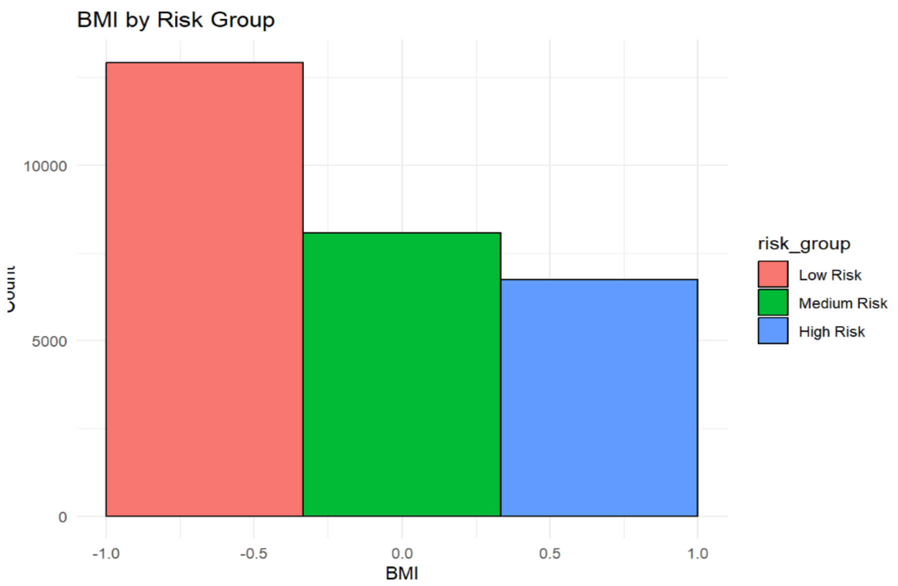
From the comparison of all five models, it is evident that the **Random Forest** model achieved the best overall performance across all key evaluation metrics. It had the highest sensitivity is **0.807**, specificity is **0.873**, accuracy is **0.839**, and AUC is **0.921**, indicating its strong ability to correctly identify both positive and negative cases of heart disease. While models like XG-Boost and Neural Networks also performed well, particularly in terms of AUC and accuracy, Random Forest consistently outperformed them. Therefore, it can be concluded that the Random Forest model is the most effective and reliable choice for heart disease prediction in this study.

Further data models are run using optimization models and finding models that outperforms and give the better results.

Clustering

Clustering method is used in data analysis to group similar data points together based on certain features or patterns. The main goal is to make sure that data points within the same group are more alike to each other than to those in other groups. Gower distance is a way to measure how different data points are, especially when the dataset has both numbers and categories (like age and gender). After calculating these differences, we can use hierarchical clustering with the Ward method to group similar data points together.

The random forest model is used here to predict probabilities of heart disease. Predicted probabilities are grouped into risk categories: Low Risk, Medium Risk, and High Risk. The data is summarized by risk groups, computing average and standard deviation for numeric features.



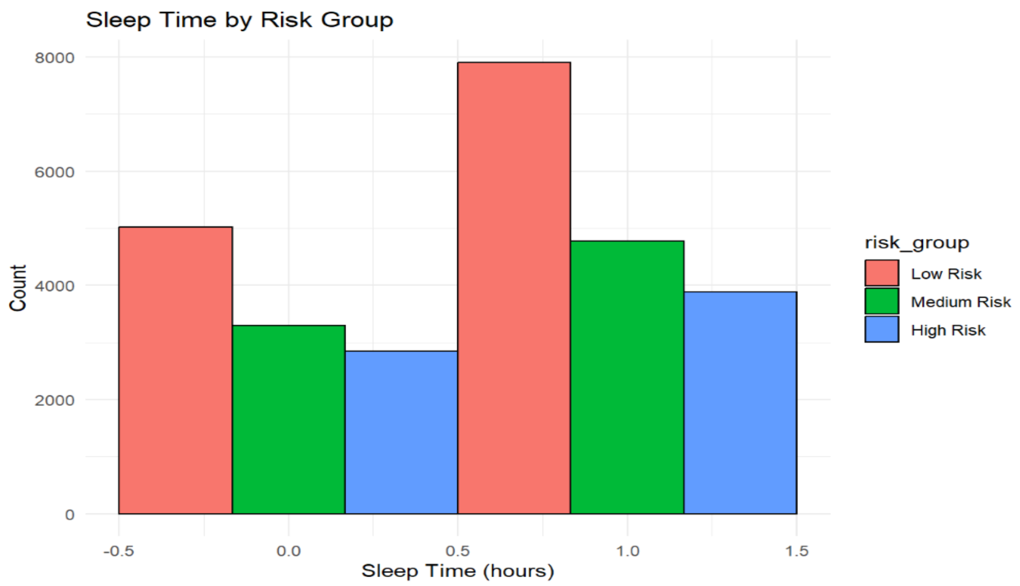
- BMI Distribution:

Low-risk group makes up the largest share of the cohort. Medium and High-risk groups shrink successively, showing an inverse relationship between BMI-related risk and participant count.

Similar graphs are plotted for the below variables,

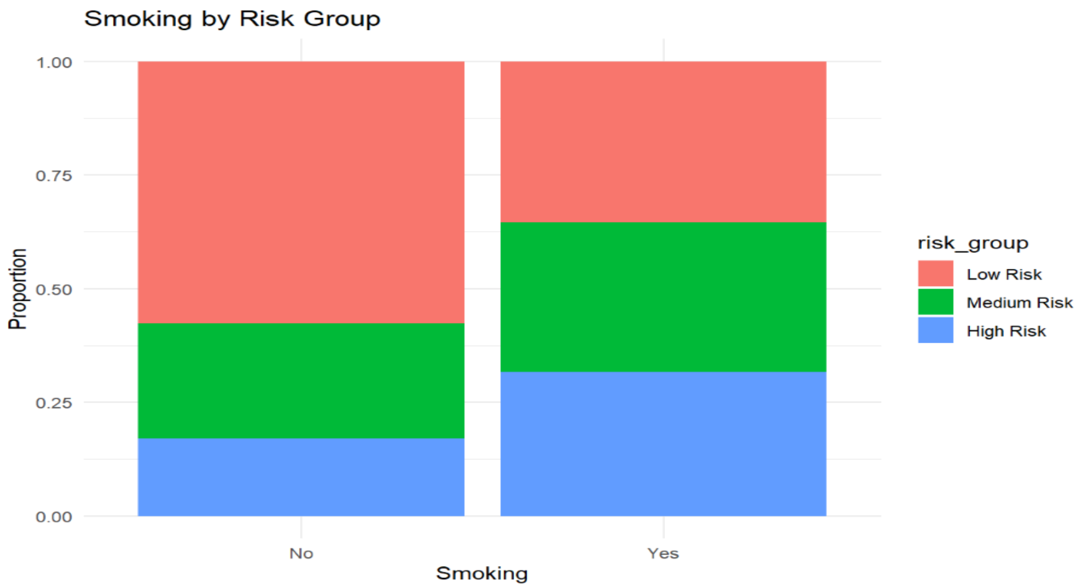
- Sleep-Time Distribution:

The graph indicates that individuals in the low-risk group predominantly have moderate sleep times, while medium and high-risk groups have more evenly distributed sleep durations. This suggests that extreme sleep times (too short or too long) might be associated with increased heart disease risk.



- Smoking and Risk Group Distribution:

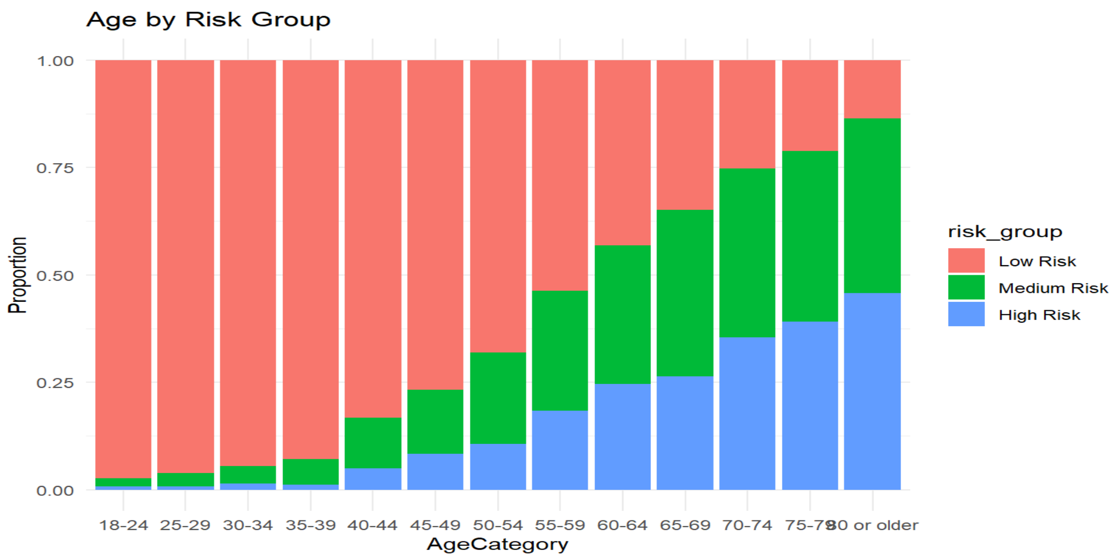
  This stacked bar chart shows a higher proportion of smokers in medium and high-risk categories compared to non-smokers, emphasizing that smoking could be a significant factor contributing to heart disease risk.



- Age Category vs. Risk Group:

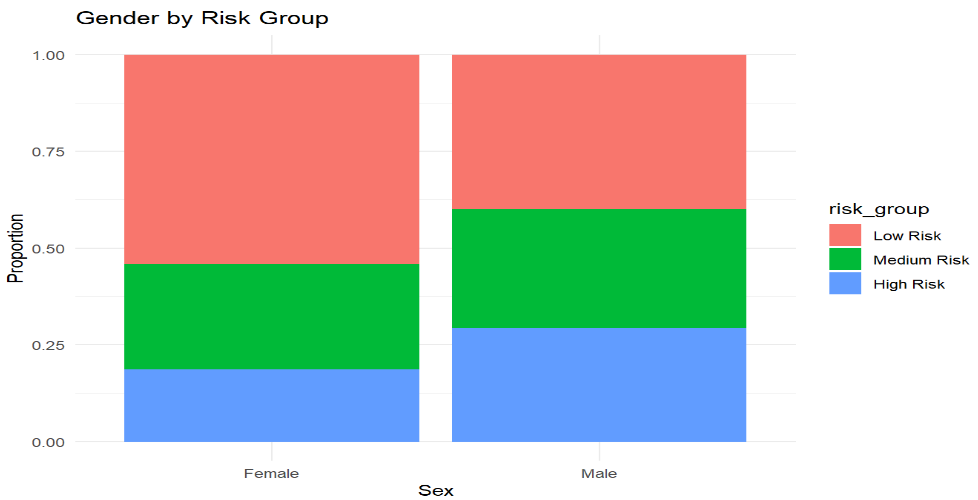
Age plays a critical role in determining heart disease risk. Individuals aged 18 to 44 generally fall into the low-risk category, while those between 45 and 64 begin to show a noticeable increase in medium to high risk. For individuals aged 65 and above, the risk is predominantly high. This clear age-wise progression highlights aging as a major risk factor.

Therefore, promoting healthy lifestyle changes during middle age is essential to reducing the likelihood of heart disease in later years.



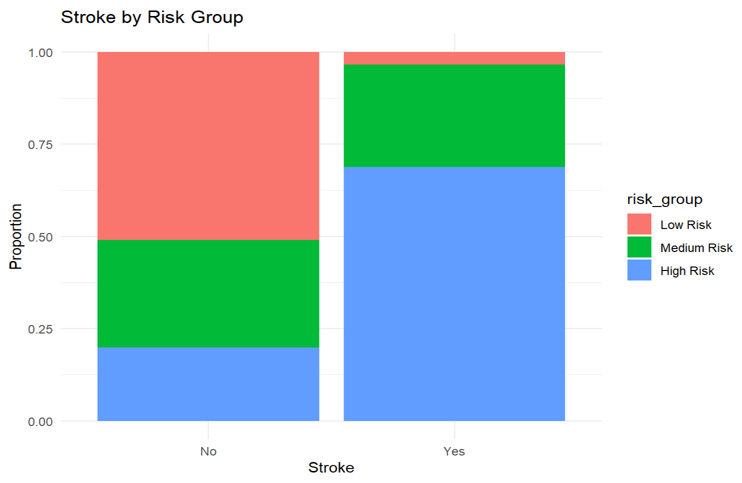
- Gender Distribution:

Risk distribution shows subtle gender-based differences in heart disease likelihood. Females tend to fall more frequently into the low-risk category, whereas males show a higher concentration in the medium- and high-risk groups. While the difference is not drastic, it highlights a consistent pattern, suggesting that men may benefit from more frequent screening and proactive health interventions to mitigate future risk.



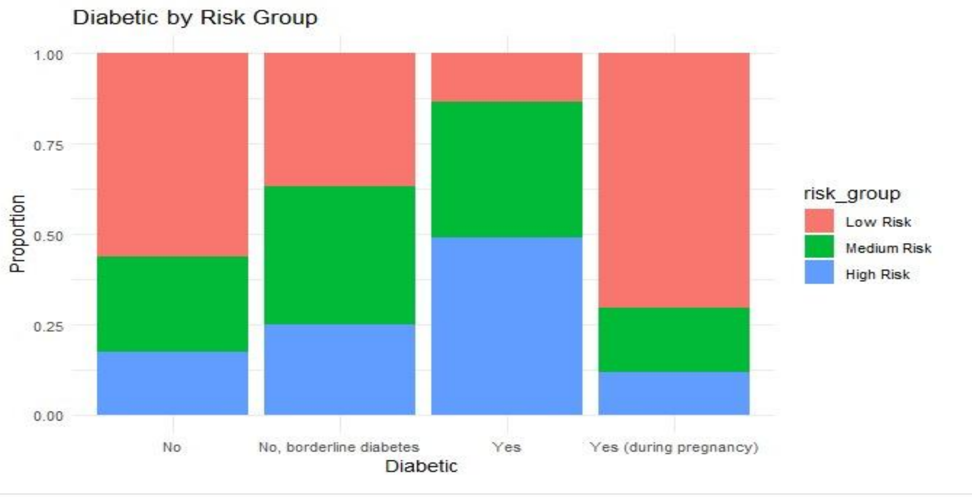
- Stroke vs. Risk Group:

The relationship between stroke history and heart disease risk is notably significant. Individuals with no history of stroke are largely found in the low- to medium-risk categories, whereas those with a prior stroke fall predominantly into the high-risk group. This pattern reinforces that a previous stroke acts as a strong clinical indicator for cardiovascular vulnerability, necessitating more intensive monitoring, early intervention, and lifestyle management to mitigate further complications.



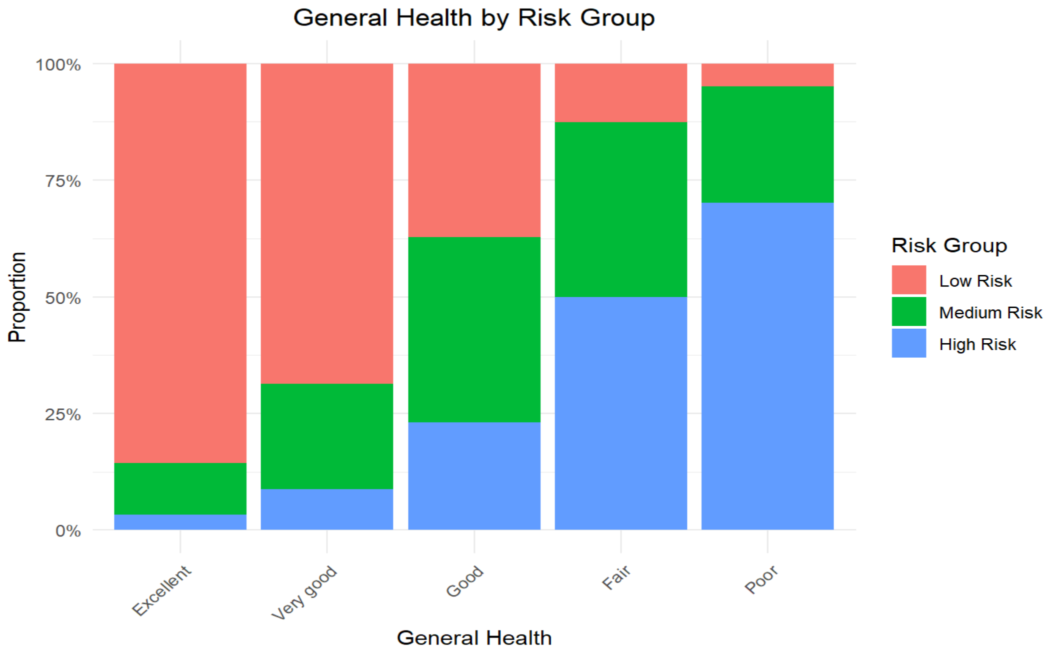
- Diabetic Status vs. Risk Group:

Diabetes status has a strong influence on heart disease risk. Individuals without diabetes are mostly in the low-risk category, while those with borderline or diagnosed diabetes show higher chances of being in the medium to high-risk groups. Particularly, diagnosed diabetics fall predominantly into the high-risk group. Managing blood sugar effectively is essential to reduce heart-related complications.



- General Health Distribution:

The bar chart shows the distribution of health risk groups across different self-reported general health levels. As general health declines from "Excellent" to "Poor," the proportion of individuals in the High-Risk group (blue) increases significantly. Conversely, those reporting "Excellent" or "Very good" health are mostly in the Low-Risk group (red).



Recommendations

Based on the predicted risk categories, tailored health recommendations were proposed to guide individuals toward healthier outcomes. For those in the high-risk group, immediate lifestyle modifications such as strict dietary changes, smoking and alcohol cessation, regular medical supervision, and possibly medication are advised. Medium-risk individuals are encouraged to adopt a heart-friendly diet, engage in daily physical activity, and avoid harmful habits. Low-risk individuals should continue their current healthy habits with regular check-ups to maintain their well-being. These personalized guidelines aim to reduce risk and promote long-term cardiovascular health.

Conclusion

This study effectively designed and assessed various machine learning models to predict the risk of heart disease by utilizing a rich dataset encompassing health indicators, behavioral habits, and demographic attributes. Among all the models tested, the Random Forest algorithm emerged as the top performer, achieving superior accuracy, sensitivity, specificity, and AUC. Its ability to handle complex interactions between features and provide robust predictions highlights its suitability for medical risk classification tasks.

The success of this project underscores the potential of machine learning in transforming preventive healthcare. By enabling early detection of individuals at higher risk, such predictive models can guide timely interventions, personalized health planning, and efficient resource allocation. Ultimately, this approach promotes a shift from reactive treatment to proactive disease prevention, which is crucial in minimizing the long-term burden of cardiovascular conditions on individuals and healthcare systems.

Future Work

Future research can focus on enhancing model performance by experimenting with advanced deep learning techniques, such as convolutional or recurrent neural networks, to better capture complex patterns in health data and integrating real-time data. Inclusion of longitudinal data to observe changes in patient health over time, enabling early detection of deteriorating conditions.

There is also potential to deploy the predictive model as a clinical decision support tool through a user-friendly web or mobile application that healthcare professionals can use in real-world settings.

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